



Introduction to Macroeconomics

Course by Prof. Marcelo Olarreaga

Teaching Assistant: Utsoree Das

Solution: Problem Set 4

2025

Solution 1

1. The equilibrium of the market of money requires that real money supply equals to money demand, where $\left(\frac{M}{P}\right)^d$ is the money demand and $\frac{M}{P}$ is the money supply

$$\left(\frac{M}{P}\right)^d = \frac{M}{P}$$

$$cY = \frac{M}{P}$$

Using quantitative equation we can write the currency

$$MV = PY$$

where M is the money supply, V is the velocity of money, P is the price level and Y is the quantity of output. V is the speed at the money changes hands (the average number of times that any piece of money is used for transactions in a given period of time).

By dividing the last two equations above, we find that $c = 1 / V = 0.5$.

Therefore, if V is the average number of times that any piece of money is used for transactions in a given period of time, c can be interpreted as the average amount of time it takes for any piece of money to be used for one transaction.

2. The equation of the quantitative theory of money can be rewritten in terms of growth rates :

$$MV = PY$$

$$\ln(MV) = \ln(PY)$$

$$\ln M + \ln V = \ln P + \ln Y$$

$$\frac{\partial(\ln M)}{\partial t} + \frac{\partial(\ln V)}{\partial t} = \frac{\partial(\ln P)}{\partial t} + \frac{\partial(\ln Y)}{\partial t}$$

$$\frac{\partial(\ln M)}{\partial M} \frac{\partial M}{\partial t} + \frac{\partial(\ln V)}{\partial V} \frac{\partial V}{\partial t} = \frac{\partial(\ln P)}{\partial P} \frac{\partial P}{\partial t} + \frac{\partial(\ln Y)}{\partial Y} \frac{\partial Y}{\partial t}$$

$$\frac{\frac{\partial M}{\partial t}}{M} + \frac{\frac{\partial V}{\partial t}}{V} = \frac{\frac{\partial P}{\partial t}}{P} + \frac{\frac{\partial Y}{\partial t}}{Y}$$

$$\frac{\dot{M}}{M} + \frac{\dot{V}}{V} = \frac{\dot{P}}{P} + \frac{\dot{Y}}{Y}$$

where $\dot{X} = \frac{\partial X}{\partial t}$ is the velocity of variation of X , that is, by how much X changes per unit of time. Therefore, $\frac{\dot{X}}{X}$, or the relative change of X (with respect to the total amount of X) is the growth rate of X . Note that $\frac{\dot{M}}{M}$, $\frac{\dot{V}}{V}$ and $\frac{\dot{Y}}{Y}$ are the growth rates of money, velocity of money and output (or real GDP), respectively. $\frac{\dot{P}}{P}$ is the growth rate of the prices or, in other words, inflation.

3. Writing the equation for the quantitative change in terms of percentage changes, we have : $\Delta\%M + \Delta\%V = \Delta\%P + \Delta\%Y$. Given that inflation is defined as the percentage change in price and the speed of money is constant over time, we infer :

$$\Delta\%M = \pi + \Delta\%Y$$

$$\pi = \Delta\%M - \Delta\%Y$$

So, the growth rate of 2 % of the money would result in a zero inflation.

Solution 2

Answer (b)

The government finances public spending by levying taxes, such as income and sales taxes, by borrowing from the public, by selling government bonds or also by printing money. When the government raises revenue by printing the money it needs, it is said to levy an inflation tax : by printing money, the government increases the money supply, the price level increases, and therefore the money held by any economic agent is able to buy less. Since by printing money it is as if the government was levying a tax on the money held by economic agents, this way of financing public spending is called inflation tax.

When a government has high public spending and an insufficient tax revenue, it ends up running a government budget deficit. If it cannot increase taxes and it has a limited ability to borrow, the only way to reduce the budget deficit is by printing money. In such a situation, the government might be tempted to print money to reduce its budget deficit, regardless of the risk of inflation (or hyperinflation). Thus, (b) is correct. The economists and the government know these risks, and therefore (a) is false.

Inflation generates costs and distortions in the economy (Mankiw, pp. 621-629) and therefore (c) is false.

Finally, an inflation tax means that the more money you hold, the higher is the tax you pay. As a result, it could be considered a sort of progressive tax. Even if we agreed to define “equitable” as a redistribution of money from the richer to the poorer, the inflation tax is a transfer of revenue from the households to the government. A well-designed progressive tax followed by a redistribution of money to the poorer households would be, in this sense, more “equitable”. Thus, (d) is false.

Solution 3

Since there are no savings in country A, the national savings is equal to public savings and open economy, we have :

$$S_g^a = I^a + NX^a = 50 - 30 = 20$$

Since everything exported by country B is imported into country A. We deduce therefore a surplus of 20, where

$$T^a - G^a = 20 ; G^a = 30$$

We can therefore deduce the GDP of country A, which is given by the identity accounting in an open economy :

$$GDP^a = C^a + I^a + G^a + NX^a = 200 + 50 + 30 - 30 = 250$$

Given that the global investment equals global savings, we can write :

$$S^w = I^w = 0.2(GDP^a + GDP^b) = 0.2 * (550) = 110$$

Investment in country B will be :

$$I^b = 110 - I^a = 60$$

with private savings equal to :

$$s_p^b = I^b + NX^b = 60 + 30 = 90$$

We therefore deduce public spending from national identity :

$$GDP^b = C^b + I^b + G^b + NX^b ; G^b = 300 - 180 - 60 - 30 = 30$$

We finally see that the correct answer is (c)

Solution 4

Answer (b)

The Big Mac index is used to find the **theoretical exchange rate** between the USD and CHF by comparing the price of hamburger in the two countries. It is given by :

$$e_{th} = \frac{P_{CHF}^{BM}}{P_{USD}^{BM}}$$

If $e_{obs} > e_{th}$, then it is expected that the CHF would appreciate vis-a-vis the USD (and the USD depreciate vis-a-vis the CHF).

First, calculate the theoretical exchange rates of the three currencies against the USD. Then we compare their exchange rates observed in the market.

1) Japan :

$$e_{th}^{YEN} = \frac{P_{YEN}^{BM}}{P_{USD}^{BM}} = \frac{270}{3} = 90$$

Accordingly, $e_{obs}^{YEN} > e_{th}^{YEN}$ ($e_{obs}^{YEN} = 121$), we therefore expect an **appreciation** of the Yen against the USD.

2) China :

$$e_{th}^{YUAN} = \frac{P_{YUAN}^{BM}}{P_{USD}^{BM}} = \frac{12}{3} = 4$$

Therefore, $e_{obs}^{YUAN} > e_{th}^{YUAN}$ ($e_{obs}^{YUAN} = 7.8$), we therefore expect an **appreciation** of the Yuan against the USD.

3) Switzerland :

$$e_{th}^{CHF} = \frac{P_{CHF}^{BM}}{P_{USD}^{BM}} = \frac{6}{3} = 2$$

Thus, $e_{obs}^{CHF} < e_{th}^{CHF}$ ($e_{obs}^{CHF} = 1.25$), we therefore expect an **depreciation** of CHF against the USD.

With this information, we can assert that the answer a) is false because it is expected a **depreciation** of CHF against the USD. The answer c) is false because the Yen and Yuan is undervalued with respect to USD (because it expects an **appreciation** of these currencies against the USD). The answer d) is false for the same reason. The answer b) is correct, the CHF is overvalued because its theoretical exchange rate is higher than the observed exchange rate.

Solution 5

Answer (a)

Relative PPA Theory states that movements in the exchange rate between the currencies of two countries reflect the difference in inflation rates of these countries. In direct quotation (we take Switzerland as domestic country) :

$$\Delta\%E = \Delta\%P - \Delta\%P^* = \pi - \pi^*$$

In our case :

$$\Delta\%E = 2\% - 8\% = -6\%$$

The exchange rate $E = \frac{CHF}{YUAN}$ is expected to decrease, therefore we expect an **appreciation of the CHF**.

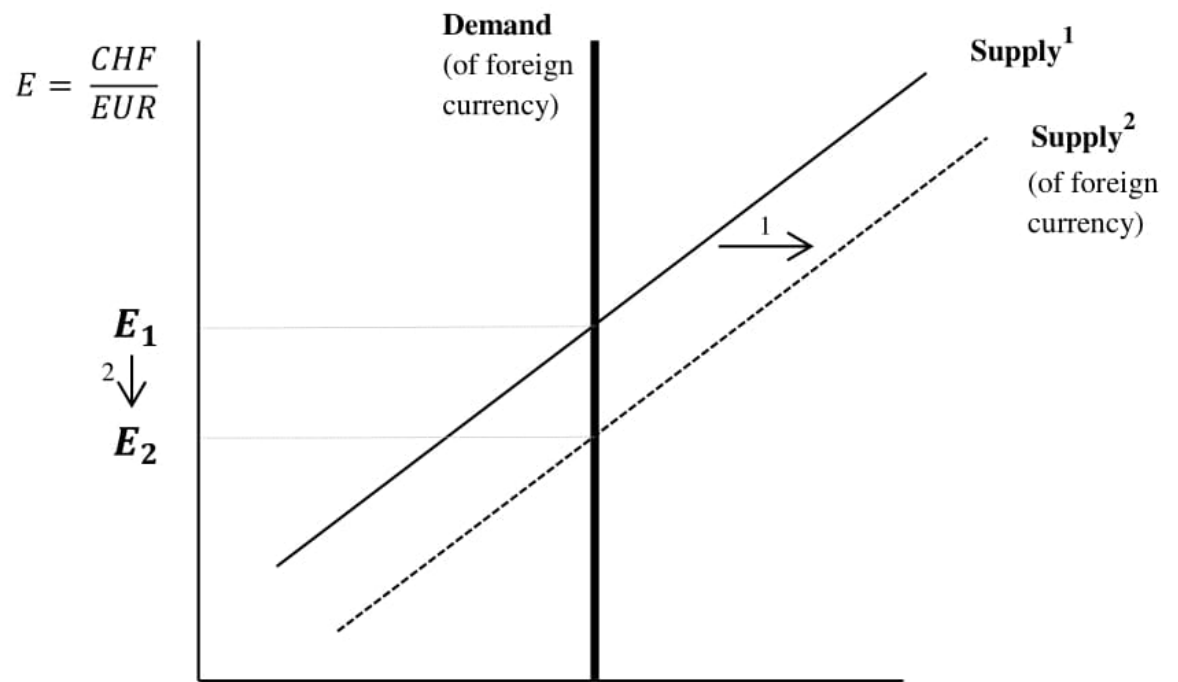
Solution 6

1. What is the impact on the demand for Swiss francs on the currency market?

When an exogenous preference shock leads to the French liking Swiss cheese more, this causes a positive impact on Swiss net exports. To pay for these additional imports from Switzerland, the French need to sell Euros to buy Swiss francs. The supply of Euros increases, shifting the supply curve to the right in our chart. The graph shows the market of Euros with on the y axis, the exchange rate E in terms of CHF/EUR (domestic currency / foreign currency), and on the x axis, the amount of Euros on the market. The supply shift to the right causes CHF appreciation.

Note that the demand (represented by a vertical line) does not change, as it is determined by the Net Capital Outflow (NCO) which depends on the real interest rate. Since nothing in the question indicates that real interest rate has changed, the NCO remains constant, and therefore the demand for EUR remains unchanged as well.

To avoid confusion with the graphical representation in some books, note that in some books the y axis is expressed with indirect quotation (in terms of $E = \frac{EUR}{CHF}$), while the x axis is expressed in terms of quantity of *domestic* currency (whereas in our graph, the x axis represents the quantity of *foreign* currency). Since the demand of foreign currency is equivalent to the supply of domestic currency (every EUR bought must be matched to every CHF sold), the vertical line which we call the *demand* of foreign currency, is often called the *supply* of domestic currency by some books. Similarly, what is for us a *supply* of foreign currency, in some books is called the *demand* of domestic currency. In addition, since we are using direct quotation instead of indirect, our supply curve is upwards sloping (whereas in some books that use indirect quotation, this same line is downwards sloping). It is up to you which representation you wish to use, the answers are equivalent in both cases.



2. What is the impact on the value of the Swiss franc on the foreign exchange market?

As we have seen, the rightward shift in supply causes a decrease in the exchange rate which is expressed in terms of domestic currency / foreign currency. Following this decrease, one Swiss franc can buy more foreign currency. The Swiss franc is appreciated (its relative value has increased).

3. What's happening on net exports and net capital outflow?

Following the appreciation of the Swiss franc, the Swiss goods become relatively more expensive for foreigners. Therefore, net exports decrease until they reach their original level. How do we know that net exports should return to their original value? As you have seen in class :

$$NX(E) = NCO(R)$$

meaning that NCO (net capital outflows) depends only on the interest rate R , and net exports (NX) and outflows must be balanced. Since the interest rate R has not changed (we have no indication of this), NCO has not changed. Therefore, net exports must adjust and return to their initial value (before the fashion of Swiss cheese in France).

Additional exercises - Not covered in seminar

Solution 7

Answer (a)

Solution 8

1. If the Chinese decided that they no longer wanted to buy U.S. assets, what would happen in the US market for loanable funds? What would happen to the US interest rates, the US savings and the US investment?

If Chinese stops buying U.S. assets, the net outflow of capital from US will grow. This “capital flight” means that investors need loanable funds to invest outside US, therefore the demand for loanable funds to US increases (demand from D_1 to D_2). In the graph, we can see the increase in net capital outflow (NCO moves to the right) and the rightward shift of the demand in the market for loanable funds. It has results :

1. Higher U.S. interest rates (increases from r_1 to r_2).
2. An increase in savings, because the equilibrium interest rate increases.
3. A drop in U.S. investment. When there is a flight of capital from one country, investments fall. Then investors prefer to put their capital outside the country.

2. What would happen in the market for foreign currency exchange? In particular, what would happen to the value of the USD and the US trade balance?

The increase in net capital outflow (NCO) causes an increase in the supply of dollars (because the dollars must be exchanged against another currency to invest outside US). In the graph the supply of dollars moves to the right of S_1 to S_2 . The dollar depreciates then (the exchange rate expressed in USD / foreign currency increases), and it becomes relatively cheaper for foreigners to buy U.S. goods. Net exports increase (and therefore the trade balance as well). In summary :

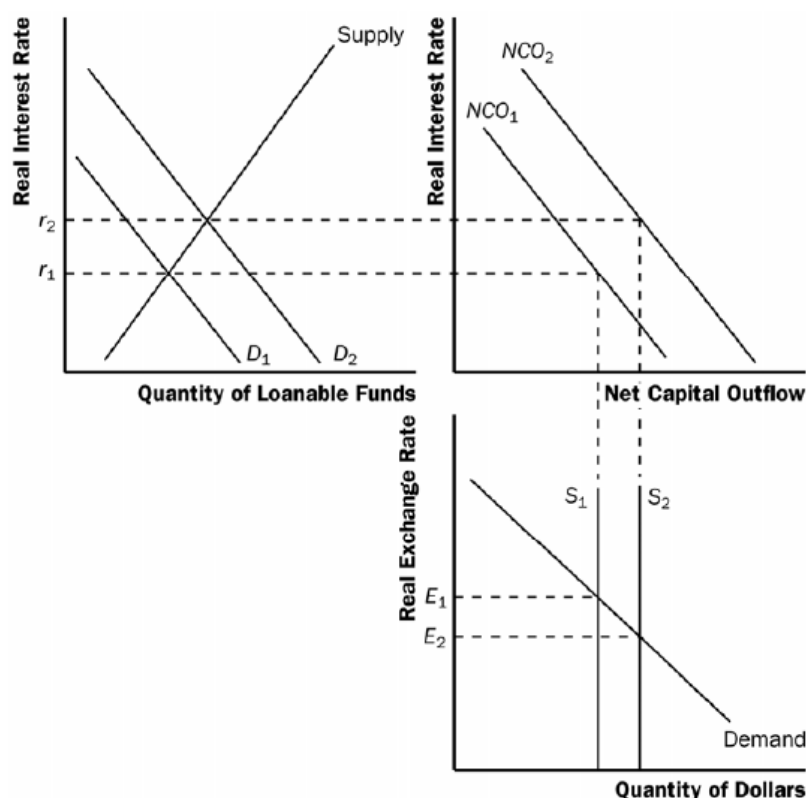
1. The dollar depreciates.
2. The trade balance improves.

The currency market being represented is the market of foreign currency and foreign exchange rate is expressed in terms of domestic currency / foreign currency. Why? When the “good” in the market is the foreign currency, we have the y-axis as the price of the foreign currency, that is to say the amount of domestic currency that must be given to receive one unit foreign currency : one unit of E is the amount of USD per unit of foreign currency. If E

increases, the dollar depreciates (we need more dollars to buy one unit of foreign currency). Capital flight from the domestic country causes an increase in the demand for foreign currency.

In this exercise, we focus on the situation of the United States. We treat the problem by considering the U.S. as the domestic country. However, unlike the graphics seen in class, it shows, in the chart below, the market for domestic currency (USD) and not for foreign currency. This approach is equivalent to the one in class, but the “good” in the foreign exchange market is the dollars. Therefore, the y-axis was the price of the domestic currency, that is to say the amount of foreign currency should be spent for 1 USD. Therefore, the units E in the graph represents the amount of foreign currency / USD. If E increases, the dollar appreciates (we need more foreign currency for the dollar, the dollar rises in value). On the other hand, if E decreases, the dollar depreciates. Thus, an increase in the supply of dollars causes a decrease of E from E_1 to E_2 , which means that the dollar depreciates.

Finally, it should be noted that in the foreign exchange market (chapter “Balance open economy”), we saw that the “capital flight” increases **the demand** of foreign currency. In the chart below, we can see that in the market for domestic currency, the “capital flight” increases **the supply** of domestic currency.



Attention : graph in indirect quotation !

Solution 9

1. We know that (11 slides, open economy chapter) :

$$\begin{aligned}\text{Balance of Payments} &= \text{Current Account Balance} - \text{Net Capital Outflow} \\ &\quad - \text{Changes in Net Official Reserves} \\ &= \text{Current Account Balance} + \text{Capital Account Balance} \\ &\quad - \text{Changes in Net Official Reserves}\end{aligned}$$

Concerning the Current Account Balance : for every export of Pecunia to the rest of the world, there is an exchange of foreign currency into Pecunian currency in order to pay Pecunian producers or exporters for the goods exported. For every import from the rest of the world to Pecunia, there is an exchange of Pecunian currency into foreign currency to pay foreign producers or importers for the goods imported. Therefore, exports generate an increase in the demand for Pecunian currency and imports generate an increase in the supply for Pecunian currency. As a result, the Current Account Balance is also called "net exports", because it contains the Trade Balance = Exports - Imports, reflects the excess demand for Pecunian currency generated by the international flows of goods that involve Pecunia.

Concerning the Capital Account Balance : when foreigners want to invest in Pecunia and buy Pecunian assets (capital inflow to Pecunia), they need to exchange their foreign currency into Pecunian currency in order to buy these assets, which generates an increase in the demand of Pecunian currency. When Pecunians want to invest abroad and buy foreign assets (capital outflow from Pecunia), they need to exchange Pecunian currency for foreign currency, which generates an increase in the supply of Pecunian currency. Therefore, Capital Account Balance, which is equal to the capital inflow to Pecunia-capital minus the outflow from Pecunia, or the net capital inflow to Pecunia, reflects the excess demand for Pecunian Currency generated by the international flows of capital that involve Pecunia.

Concerning the Changes in Net Official Reserves : when the Pecunian central bank increases its reserves of foreign currency, the change in the net official reserves is positive. To that end, the Pecunian Central Bank needs to buy foreign currency using Pecunian currency, which leads to an increase in the supply of Pecunian currency. When the Pecunian central bank decreases its reserves of foreign currency, it sells foreign currency and buys Pecunian currency, which leads to an increase in the demand of Pecunian currency. As a result, the Changes in Net Official Reserves reflects the excess supply of Pecunian currency generated by the change in the reserves of the Pecunian central bank, and if we place a minus sign in front of it, it reflects the excess demand of Pecunian currency generated by the change in the reserves of the Pecunian central bank.

Therefore, the Balance of Payments takes into account all the transactions that involve the exchange of Pecunian currency into foreign currency and vice versa, and is considered to represent the excess demand for Pecunian currency. When the Pecunian currency

market is at equilibrium, the excess demand for Pecunian currency generated by all these transactions must be equal to 0.

On the other hand, it should be noted that the Capital Account Balance corresponds to the net capital inflow, and therefore the NCO is equal to minus the Capital Account Balance. A positive (negative) Capital Account Balance means that there is a positive (negative) excess demand of Pecunian Currency generated by a positive (negative) net capital inflow. A positive (negative) NCO means that there is a positive (negative) excess supply of Pecunian currency generated by the positive (negative) net capital outflow.

Since the net change in official reserves is equal to 0,

$$\text{Balance of Payments} = \text{Current Account Balance} - \text{Net Capital Outflow}$$

and the Balance of Payments must be equal to 0 :

$$\text{Current Account Balance} - \text{Net Capital Outflow} = 0$$

$$\text{Net Capital Outflow} = \text{Current Account Balance}$$

In other words, if the value of the exports exceeds the value of the imports, we have a positive Current Account Balance that generates an excess demand of Pecunian currency. In such a situation, we need a positive NCO that generates an excess supply of Pecunian currency of the same amount to compensate for the positive net exports. A positive NCO means that the value of foreign assets bought by Pecunians (which generates an outflow of capital from Pecunia to buy foreign assets) must exceed the value of Pecunian assets bought by foreigners (which generates an inflow of capital to Pecunia to buy Pecunian assets).

Similarly, if the value of the imports exceeds the value of the exports, we have a negative Current Account Balance that generates an excess supply of Pecunian currency. In such a situation, we need a negative NCO that generates an excess demand of Pecunian currency of the same amount to compensate for the negative net exports. A negative NCO means that the value of foreign assets bought by Pecunians must be lower than the the value of Pecunian assets bought by foreigners.

In this exercise, there is a 1 billion euro deficit in the Current Account Balance, that is :

$$\text{Current Account Balance} = -1 \text{ billion euros}$$

which means that the value of the imports to Pecunia exceeds the value of the exports from Pecunia to the rest of the world. As a result :

$$\text{Net Capital Outflow} = \text{Current Account Balance} = -1 \text{ billion euros}$$

Therefore, the answer to (1) is that 1 billion euro deficit of the Current Account Balance has generated a decrease in the NCO of 1 billion euros. This means that there is a net capital inflow to Pecunia of 1 billion euros. By importing more than it exports, Pecunia has incurred a debt with the rest of the world, which is consistent with the fact that the rest of the world has bought Pecunian assets for a value that is 1 billion euros higher than the value of the foreign assets bought by Pecunians.

2. We have seen that the NCO is equal to minus the Capital Account Balance. Therefore :

$$\text{Capital Account Balance} = -NCO = 1 \text{ billion euros}$$